At the same time, this emphasis on the technical should not obscure the psychological, social and cultural aspects of these technologies. These technologies are reshaping the way individuals and organisations collaborate and share information. This digital convergence, in an important sense, removes, and maybe more importantly, obscures, physical and temporal boundaries that in the past served as impediments between individuals and the world.

When communication integrates, maps and borders have less meaning—if any—and barriers of distance begin to dissolve. In some ways these technologies of 'action at a distance' simplifies our lives, since it allows us to connect with the world in ways that we might not otherwise. It becomes the centre of our social network and constructs a 'virtual space' within which we live, learn and work.

Pedagogical Convergence

Living in the physical and social world—for both work and play—requires acquiring knowledge. To survive and thrive in these new social and physical spaces, new ways of learning that meet the demands of this new era will have to be developed.

The convergence of our strategies and theories of learning to live in this digitally integrated world is the second kind of convergence I am alluding to—what I call *pedagogical convergence*. This convergence deals with how our approaches to teaching and learning need to converge with the new trends in digital technologies.

Learning today can no longer be restricted to 12 + 4 years of schooling and college. Not only do we have more to learn, we need to do it more often and in less time. This puts severe stress on our existing pedagogical systems including schools, training workshops, colleges and universities.

Learning can no longer be limited to being able to recite multiplication tables by rote and such. Learning today is continuous, just in time and perpetual. This fast pace of change means that there is little time for leisure and reflection. Lifelong learning is not just an idle hope, but rather, the need of the hour.

What role can technology play in developing solutions to these problems? The first thing to remember is that education and learning is always about more than the technology. We must never forget that technology is a medium, a tool for converting ideas into action.

However, technologies do not force one set of actions over others. Technologies can be used for 'old style' learning as easily as for newer, flexible pedagogy. The technology does not care. In some sense it may even be easier to use technology to achieve 'old style' goals. This is why we often see the online course that is merely a set of static Web pages—a textbook moved onto the Web; or the naïve drill and practice programme, cosmetically enhanced by irrelevant and pointless flash and fizz. We must remember that all the technologies in the world are of no use if they are to push the same old agenda. The language of pedagogy needs to rise up to meet the language of convergence.

I remember visiting a school in Nagpur and talking to the computer teacher there. The school did not think of using computers to learn science. They did not consider using computers to develop writing skills, though there is increasing evidence that using word processors actually makes students better writers and better thinkers.

Sadly, this is true of most schools today. So the constraints are not as much cost and access, though these are critical issues—rather a failure of our collective imagination.

A part of the problem is, given the fast pace of technological change, our learning theories and pedagogical approaches have often been engaged in a game of catching up. This is not to say that we don't have a sense of what this pedagogical convergence will look like. The new pedagogical convergence is informed by thinkers such as John Dewey who urged us to base all educational strategies on the impulses of the child—the urge to communicate, inquire, construct, and express.

Some of these new understandings are inspired by the technology itself. Games and simulations, with their rich multi-player, immersive, interactive, social worlds are showing us the way to the future of learning. The title of James Gee's recent book says it all: *What Video Games Have To Teach Us About Learning And Literacy*. This is not to say that games and simulations are the only way forward. Not at all. Education is a multi-dimensional beast, and there is space for various genres and ways of doing—as long as they are thoughtfully implemented, and we resist being seduced by the shiny, sexy surfaces of technology and are prepared to peer deeper into their essences.

"Being on the verge is to be somewhere unstable and unpredictable... There is also a sense of excitement, mingled with fear, mystery and anticipation."

Deep Convergence

This brings me to the third and final kind of convergence, which for want of a better term, I call *deep convergence*. There are many people who pontificate on the social and cultural consequences of digital and pedagogical convergence. However, I take these well-meaning pronouncements with a grain of salt because forecasting often overestimates short-term effects and underestimates long-term ones.

One thing that humans do is construct new understandings of themselves, and the world they live in through using these media. As the Foresight group (http://www.foresight.gov.uk/) says in a report:

"At some point in our evolutionary past humans entered the 'cognitive niche' and fundamentally changed their relationship to the world around them and the rate at which they could develop their culture. The artificial world is about to do the same, with humans as partners in the process—the effects will be equally far reaching."